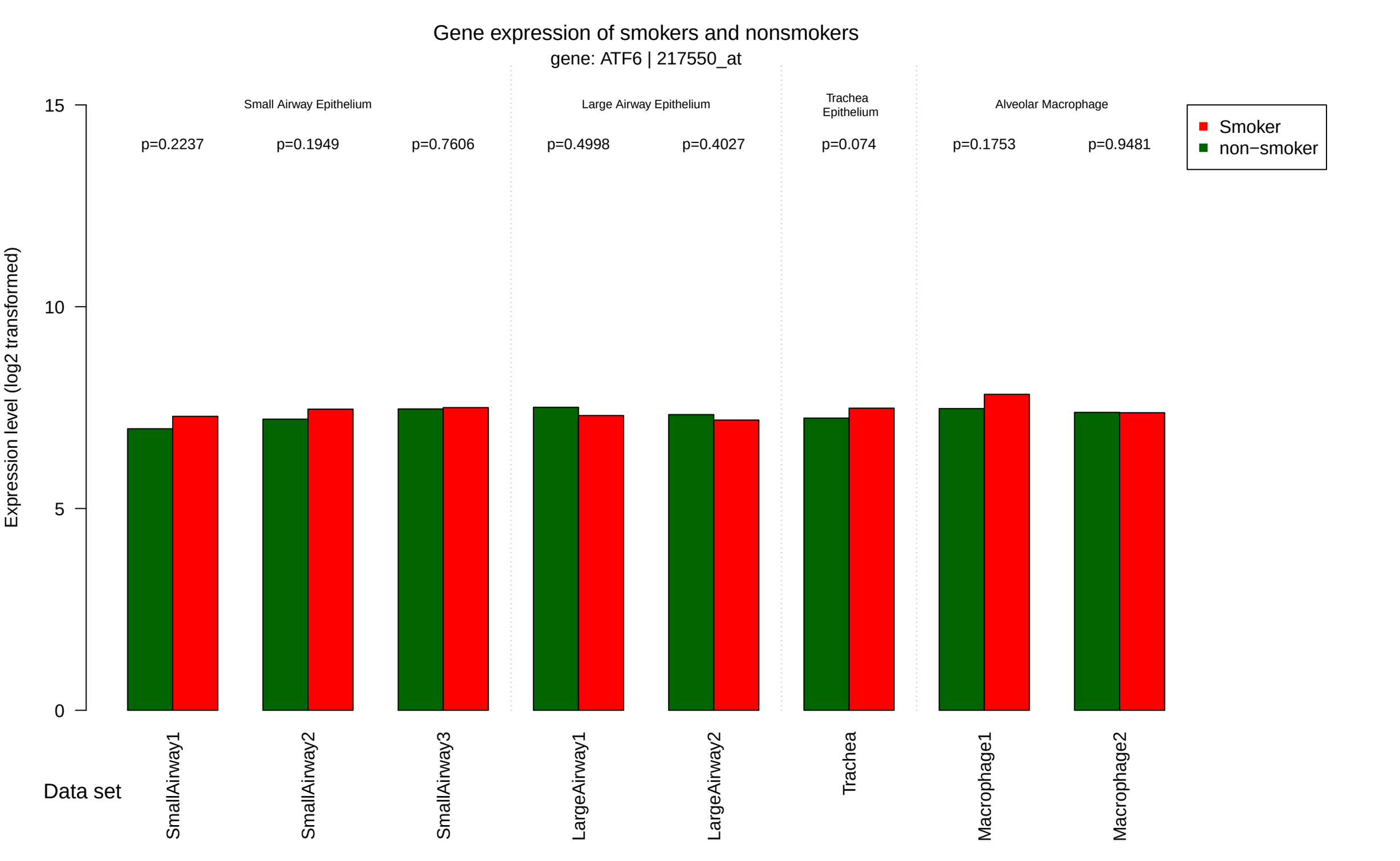
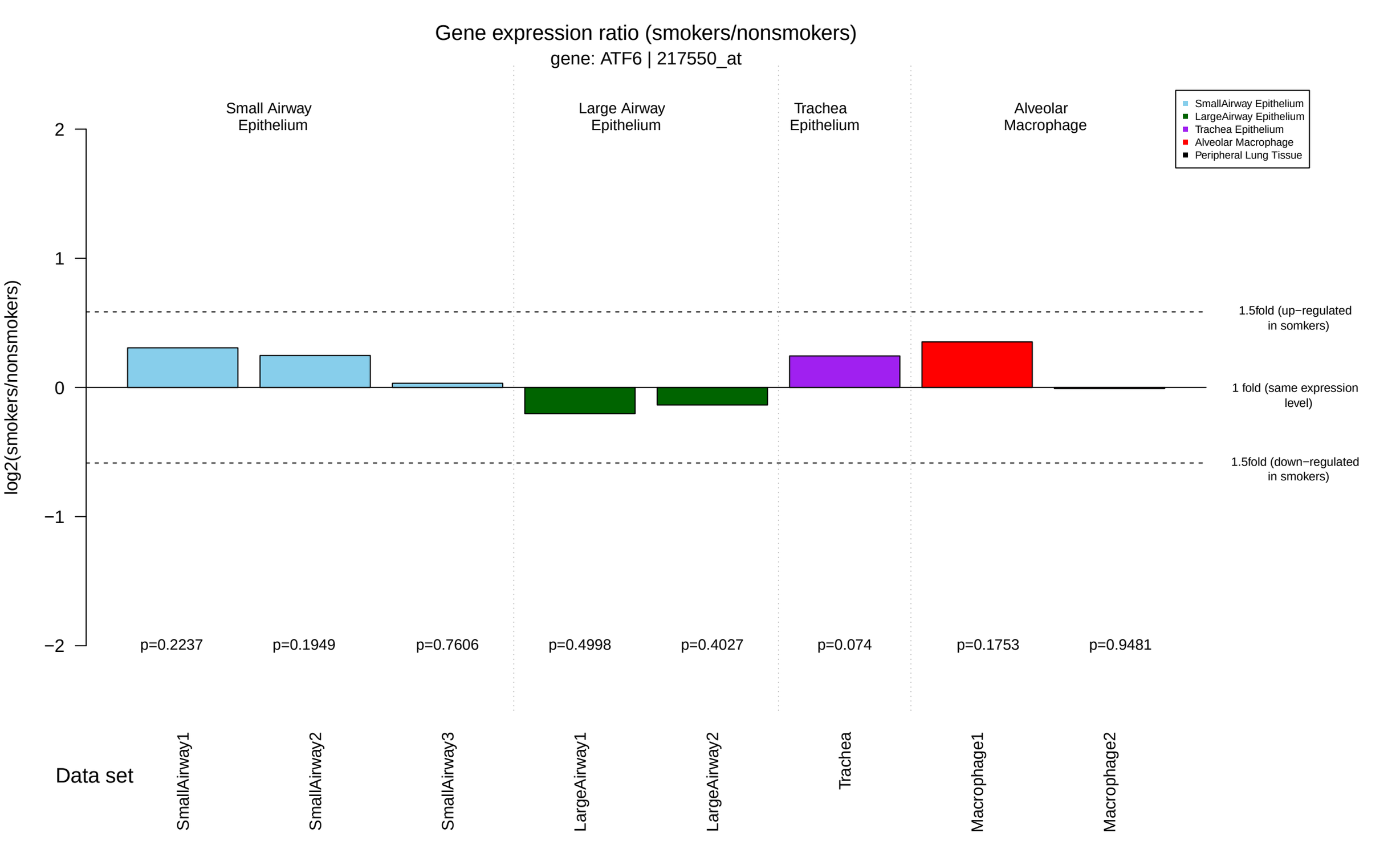
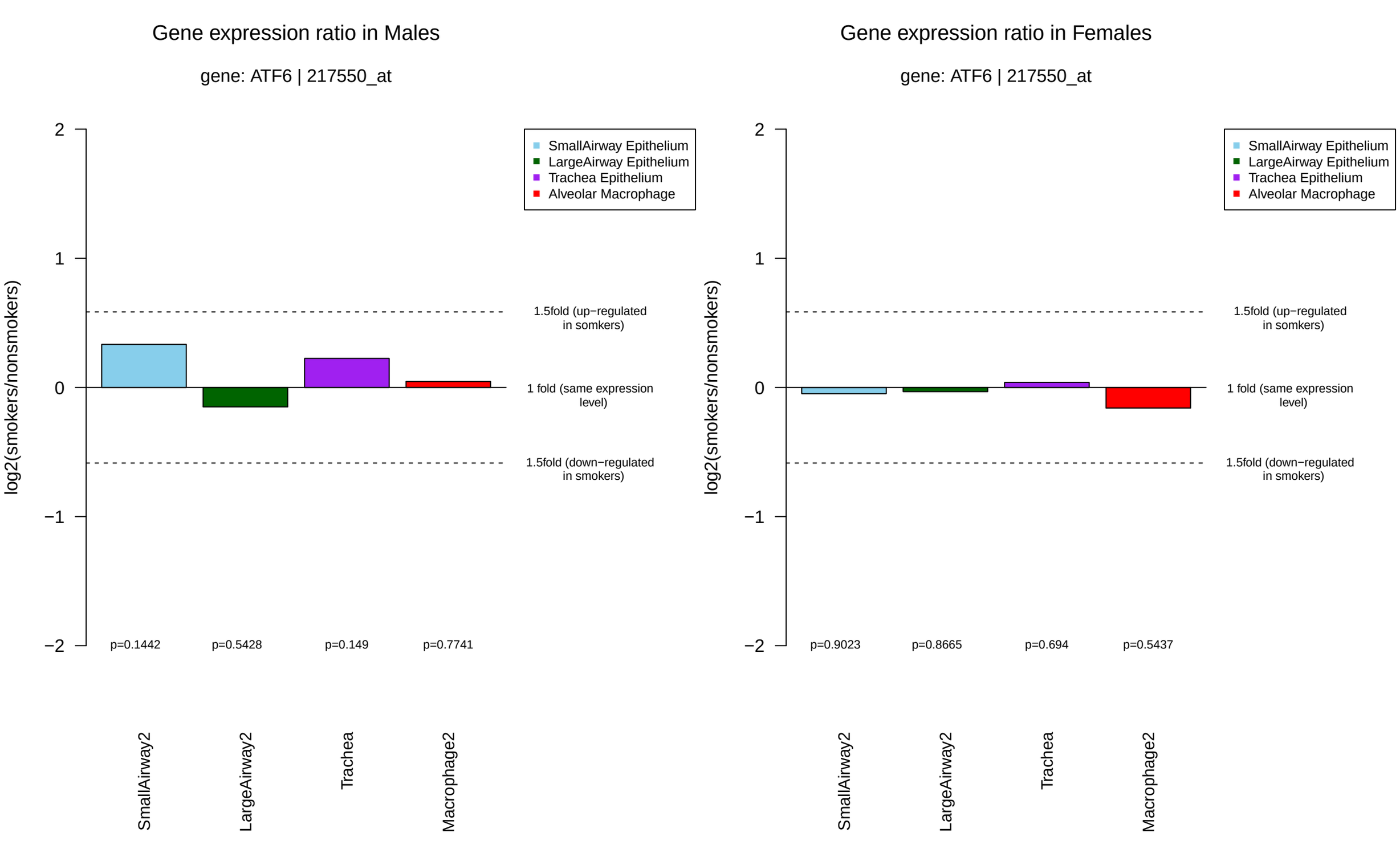
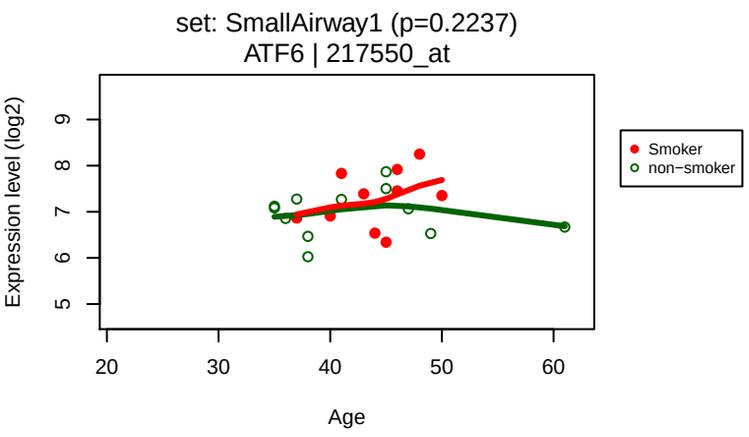
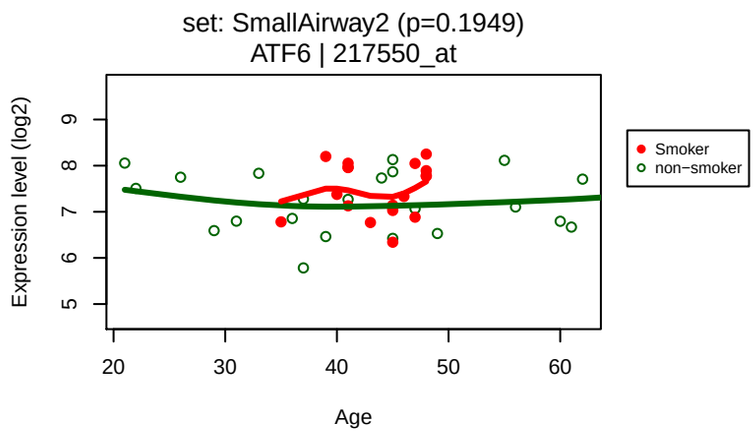
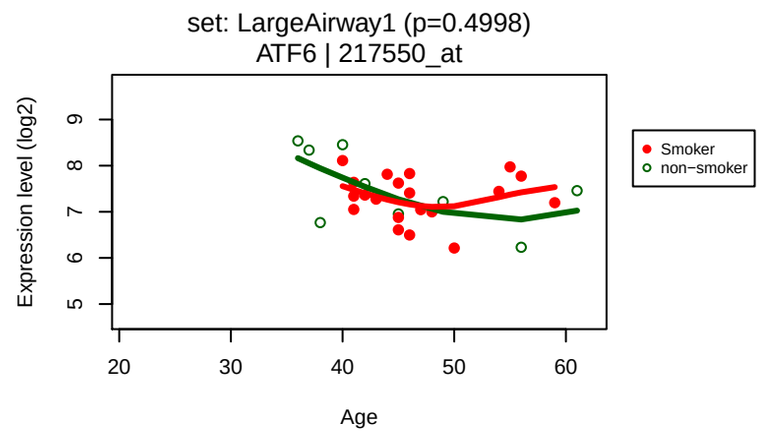

Gene expression of smokers and nonsmokers
gene: ATF6 | 217550_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2237
p=0.1949
p=0.7606
p=0.4998
p=0.4027
p=0.074
p=0.1753
p=0.9481
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

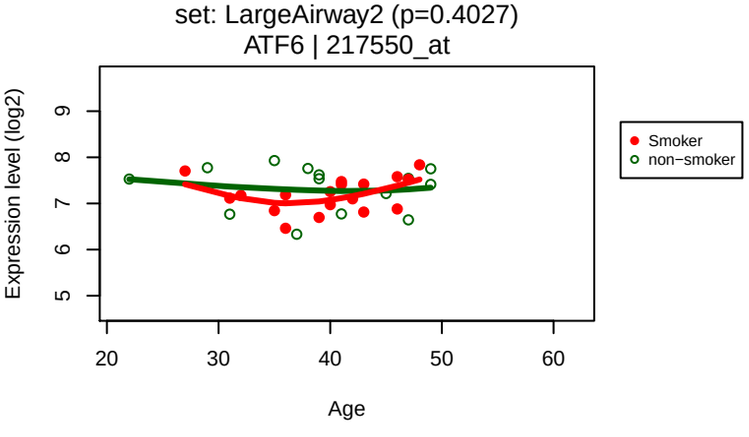

Gene expression ratio (smokers/nonsmokers)
gene: ATF6 | 217550_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2237
p=0.1949
p=0.7606
p=0.4998
p=0.4027
p=0.074
p=0.1753
p=0.9481
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

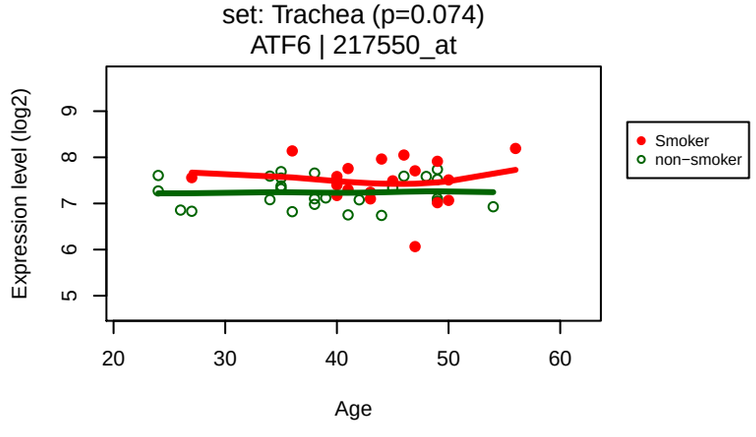

Gene expression ratio in Males
gene: ATF6 | 217550_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1442
p=0.5428
p=0.149
p=0.7741
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ATF6 | 217550_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9023
p=0.8665
p=0.694
p=0.5437
SmallAirway2
LargeAirway2
Trachea
Macrophage2

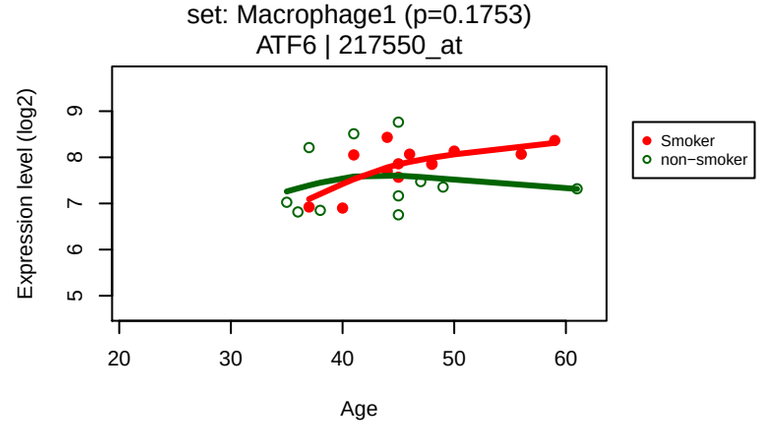

set: SmallAirway1 (p=0.2237)
ATF6 | 217550_at
Expression level (log2)
Age
Smoker
non−smoker

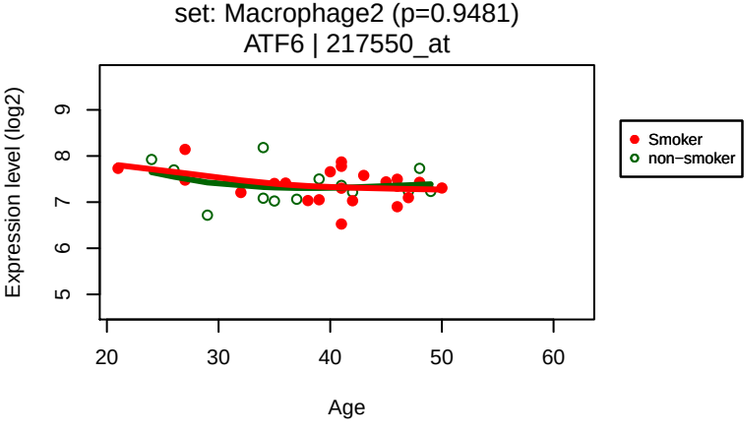

set: SmallAirway2 (p=0.1949)
ATF6 | 217550_at
Expression level (log2)
Age
Smoker
non−smoker

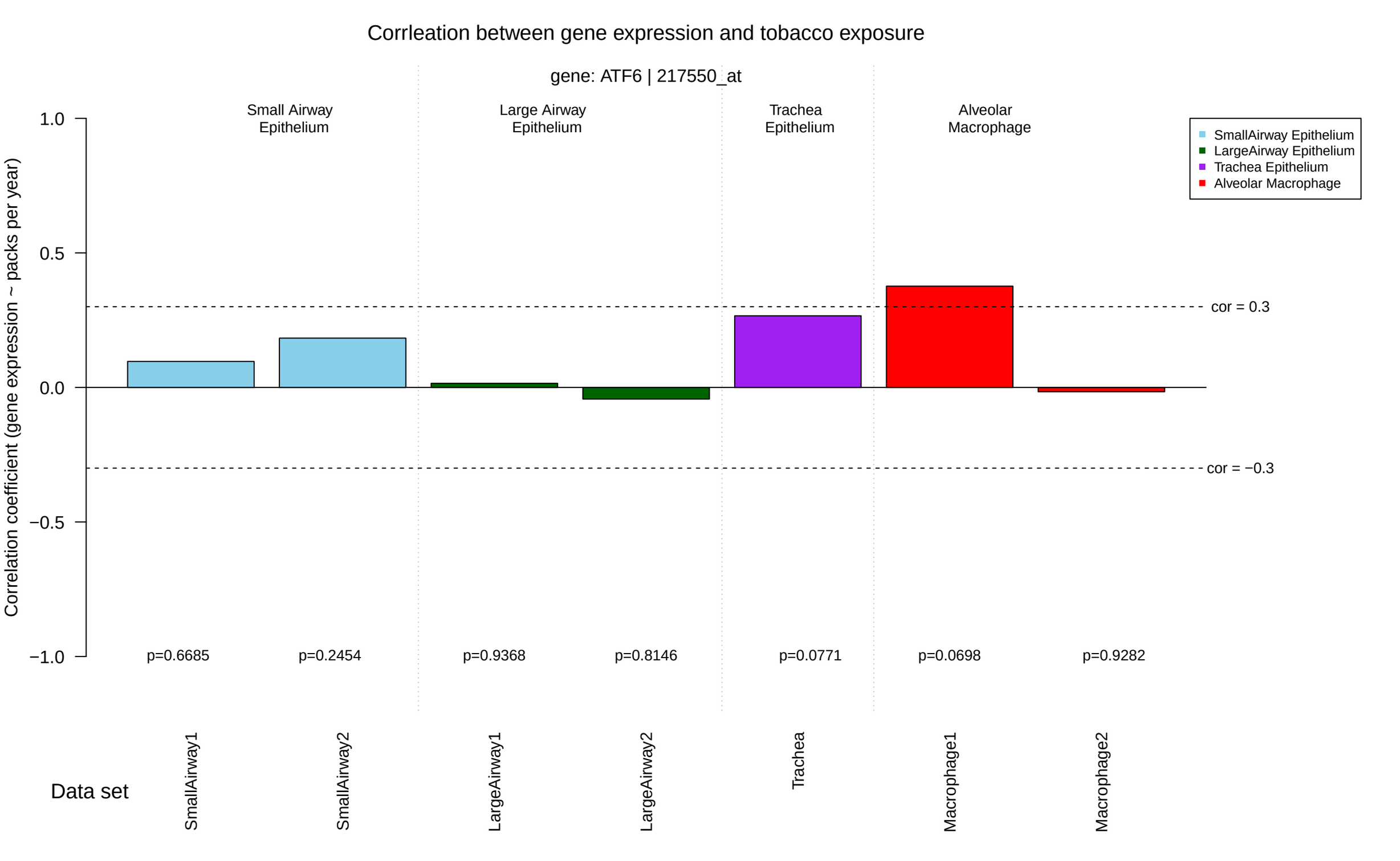

set: LargeAirway1 (p=0.4998)
ATF6 | 217550_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4027)
ATF6 | 217550_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.074)
ATF6 | 217550_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1753)
ATF6 | 217550_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9481)
ATF6 | 217550_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ATF6 | 217550_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6685
p=0.2454
p=0.9368
p=0.8146
p=0.0771
p=0.0698
p=0.9282
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2